RACHMAD FITRIYANTO

Systems Analyst | Data Science | Software & Analytics | Technical Support

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PROFESSIONAL SUMMARY

Systems Analyst and Data Science professional with nearly two decades of experience spanning IT infrastructure and technical support, systems analysis, software development, higher education, operational analysis, and applied research. Combines hands-on experience in computer troubleshooting, LAN support, network planning, and telecommunications systems with expertise in translating business and operational requirements into practical software and data solutions. Independently designed and deployed a Python/Flask and MySQL-based Internal Audit Monitoring System supporting audit monitoring across 27 branch offices. Research interests include machine learning, clustering, data mining, and explainable AI, with applied research experience in data science and model evaluation.

CORE TECHNICAL SKILLS

Data Science & Machine Learning: Python · Pandas · Data Preprocessing · Data Analysis · Data Mining · Machine Learning · Clustering · K-Means · K-Medoids · Feature Selection · Cluster Validation · Model Evaluation · Explainable AI

Systems Analysis & Software Engineering: Requirements Engineering · Business Process Analysis · Software Requirements Specification (SRS) · Data Flow Diagrams (DFD) · Entity-Relationship Diagrams (ERD) · Database Design · Software Development

Software, Database & Reporting: Flask · VB.NET · MySQL · SQL · Web Application Development · Dashboard Development · Data Visualization · Excel

IT Infrastructure & Technical Support: IT Support · LAN · Network Support · Unix/Linux · Hardware & Peripheral Support · Technical Troubleshooting · User Support

Technical Education & Research: Technical Training · Curriculum Development · Technical Documentation · Student Project Supervision · Applied Research · Academic Writing · Research Project Development

PROFESSIONAL EXPERIENCE

Internal Audit Staff | *September 2025 – March 2026*

- Conducted operational audits across 27 branch offices, reviewing customer, financial, collateral, and operational records.
- Identified limitations in the existing Excel-macro reporting workflow and proposed a simpler web-based alternative in response to operational requirements.
- Independently designed and developed a Python/Flask and MySQL-based Internal Audit Monitoring System, covering requirements analysis, database design, UI design, development, and deployment.
- Developed centralized dashboards for monitoring audit findings, status, and follow-up actions across branch offices.
- Proposed and implemented a Google Drive-based document archiving structure to address the absence of a standardized document handover system.

STMIK PPKIA Tarakanita Rahmawati | *Multiple Academic, Technical & Institutional Roles | 2019 – 2025*

Lecturer – Information Systems & Computer Science | *2020 – 2025*

- Supervised 28 undergraduate technical projects through the development lifecycle, including 14 data mining solutions and 4 CRM systems.
- Designed and delivered 8 computer science and information systems courses, supported by 8 original educational resources including a textbook, book chapters, and comprehensive modules.
- Conducted applied research in machine learning, data mining, clustering, and explainable AI, securing 2 competitive government research grants.
- Supported 7 regional community technology programs, including co-development of an automated administrative reporting solution for local educators.
- Served as an expert evaluator for 3 provincial vocational technology competitions, assessing technical competencies and project outcomes.

Secretary – Internal Quality Assurance Institute | *2024 – 2025*

- Supported institutional quality assurance through business process analysis, documentation, and coordination of internal quality activities.
- Supported operational planning and performance monitoring.

Head of Library | *2023 – 2024*

- Restructured book procurement into a needs-based process by reviewing collection requirements against course references and coordinating procurement.
- Reorganized subject classification to improve resource discoverability and alignment with academic requirements.
- Identified workload constraints within a three-person team and redistributed responsibilities to improve operational allocation.
- Redesigned monthly reporting to incorporate narrative analysis and data visualization rather than raw figures alone.
- Led service evaluation and stakeholder satisfaction surveys and managed annual work plans and library operations.

Library Staff | *2020 – 2023*

- Managed library information systems, digital documentation, and academic service quality monitoring.

IT Support & Teaching Staff | *2019 – 2020*

- Maintained campus IT infrastructure, computers, printers, and LAN; provided user and classroom technical support.
- Supported practical instruction in Systems Analysis.

IT Support Technician / Freelance Systems Analyst | *2007 – 2019*

- Progressed from telecommunications and IT support into requirements gathering, systems analysis, software development, and network planning.
- Translated client requirements into technical specifications and implementation plans.
- Independently developed software applications using VB.NET, including a room-usage and office inventory management application and a client activity/project tracking display.
- Provided onsite and remote technical consultation, troubleshooting, and internal LAN support.
- Prepared technical documentation, implementation reports, and cost estimates and coordinated with customers, vendors, and management.

SELECTED TECHNICAL PROJECT

Internal Audit Monitoring System | *Python · Flask · MySQL · Web Application · Dashboard Development*

Designed and independently developed a web-based internal audit monitoring system to replace a resource-intensive Excel-macro reporting workflow.

- Analyzed operational requirements and existing reporting workflows to identify limitations in the legacy reporting process.
- Designed the relational database structure, application workflow, user interface, and reporting components.
- Developed and deployed the application using Python, Flask, and MySQL, covering the lifecycle from requirements analysis through deployment.
- Developed centralized dashboards for monitoring audit findings, status, and follow-up actions across 27 branch offices.
- Replaced a heavy desktop-based reporting workflow with a lightweight web application for centralized monitoring.

Documentation: https://rachmadfitriyanto.pythonanywhere.com/portfolio/ncsis

DATA SCIENCE & RESEARCH

Research Focus: Machine Learning · Data Mining · Clustering · Feature Selection · Cluster Validation · Explainable AI · Data Analysis

Applied Data Science researcher with research experience spanning machine learning, data mining, clustering, feature selection, cluster validation, and explainable AI. Five of nine publications focus directly on Data Mining/Data Science, complemented by research in information security, enterprise architecture, and decision support systems. Secured 2 competitive government research grants.

- 9 publications across journals and conference proceedings
- 5 publications directly focused on Data Mining/Data Science
- 2 competitive government research grants
- 28 undergraduate technical projects supervised, including data mining projects

EDUCATION

Master of Informatics Engineering | Universitas Ahmad Dahlan | 2017 – 2019

Bachelor of Informatics Engineering | Universitas Ahmad Dahlan | 2012 – 2017

Diploma in Electrical Engineering | Universitas Negeri Yogyakarta | 2002 – 2007

LANGUAGES

Bahasa Indonesia — Native **English** — Professional Working Proficiency | B1

PROFESSIONAL LINKS

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